

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN STACK

Practical – IV

Roll No.: T010	Name: Sagar Kandari
Class: TYIT	Batch:1
Date of Assignment:15-7-26	Date/Time of Submission:15-7-26

4.React Components, Props, and State

a. Create multiple components and render them on a single page.

Code :

App.jsx

```
import Student from "../components/Student";
import Eligibility from "../components/Eligibility";
import Placementstatus from "../components/Placementstatus";
import Companyoffer from "../components/Companyoffer";
import Skillprogress from "../components/Skillprogress";
import Footer from "../components/Footer";
import "../App.css";

function App() {
  return (
    <div className="container">
      <Header />
      <Student />
      <Eligibility />
      <Placementstatus />
      <Companyoffer />
      <Skillprogress />
      <Footer />
    </div>
  );
}

export default App;
```

Skillprogress.jsx

```
function Skillprogress() {  
  return (  
    <div>  
      <h2>Skill Progress</h2>  
      <p>Java: 90%</p>  
      <p>Python: 95%</p>  
      <p>React: 85%</p>  
    </div>  
  );  
}  
  
export default Skillprogress;
```

Companyoffer.jsx

```
function Companyoffer() {  
  return (  
    <div>  
      <h2>Company Offer</h2>  
      <p>Company: TCS</p>  
      <p>Package: 7 LPA</p>  
    </div>  
  );  
}  
  
export default Companyoffer;
```

Eligibility.jsx

```
function Eligibility() {  
  return (  
    <div>  
      <h2>Eligibility</h2>  
      <p>Status: Eligible</p>  
    </div>  
  );  
}  
  
export default Eligibility;
```

Header.jsx

```
function Header() {  
  return (  
    <div>  
      <h1>M.V.L.U. College Placement Status</h1>  
      <hr />  
    </div>  
  );  
}  
  
export default Header;
```

Student.jsx

```
function Student() {  
  return (  
    <div>  
      <h2>Student Details</h2>  
      <p>Name: Sagar Kandari</p>  
      <p>Course: B. Sc. IT</p>  
      <p>Semester: V</p>  
    </div>  
  );  
}  
  
export default Student;
```

Sagar Kandari
T010

Footer jsx

```
function Footer() {  
  return (  
    <div>  
      <hr />  
      <p>© 2026 Sheth L. U. J. & Sir M. V. College</p>  
    </div>  
  );  
}  
  
export default Footer;
```

Output :

Name: Sagar Kandari

Course: B. Sc. IT

Semester: V

Eligibility

Status: Eligible

Placement Status

Status: Placed

Company Offer

Company: TCS

Package: 7 LPA

Skill Progress

Java: 90%

Python: 95%

React: 85%

© 2026 Sheth L. U. J. & Sir M. V. College

b. Pass data between components using props.

Code :

App.jsx

```
import Header from "../components/Header";
import Footer from "../components/Footer";
import HotelCard from "../components/HotelCard";
import "../App.css";

function App() {
  return (
    <div className="container">
      <Header />
      <HotelCard
        restaurant = "Domino's Pizza"
        food = "Pizza, Pasta, Beverages"
        price = {1599}
        rating = {4.5}
        discount = {20}/>
      <HotelCard
        restaurant = "McDonald's"
        food = "Burgers, Fries, Beverages"
        price = {1699}
        rating = {4.5}
        discount = {20}/>
      <HotelCard
        restaurant = "Pizza Hut"
        food = "Pizza, Pasta, Beverages"
        price = {599}
        rating = {4.5}
        discount = {20}/>
      <HotelCard
        restaurant = "Chicken Hut"
        food = "chicken Pizza, chicken hande Pasta, Beverages"
        price = {1099}
        rating = {4.5}
        discount = {20}/>
      <Footer />
    </div>
  );
}

export default App;
```

Sagar Kandari
T010

Header.jsx

```
function Header() {  
  return (  
    <div>  
      <h1>Online Food Delivery Dashboard</h1>  
      <h3>Sagar Kandari</h3>  
      <hr />  
    </div>  
  );  
}  
  
export default Header;
```

Hotelcard.jsx

```
function HotelCard(props) {  
  const discount = (props.price * props.discount) / 100;  
  const finalPrice = props.price - discount;  
  
  return(  
    <div>  
      <h2>{props.restaurant}</h2>  
      <p>Food Item : {props.food}</p>  
      <p>Original Price : ₹ {props.price}</p>  
      <p>Discount : {props.discount}%</p>  
      <p>Final Price : ₹ {finalPrice}</p>  
      <p>Rating : {props.rating}</p>  
    </div>  
  );  
}  
  
export default HotelCard
```

Footer.jsx

```
function Footer() {  
  return (  
    <div>  
      <p>© 2026 Online Food Delivery App</p>  
    </div>  
  );  
}
```

export default Footer;

Output:



c. Use the useState hook to manage and update data.

Code:

App.jsx

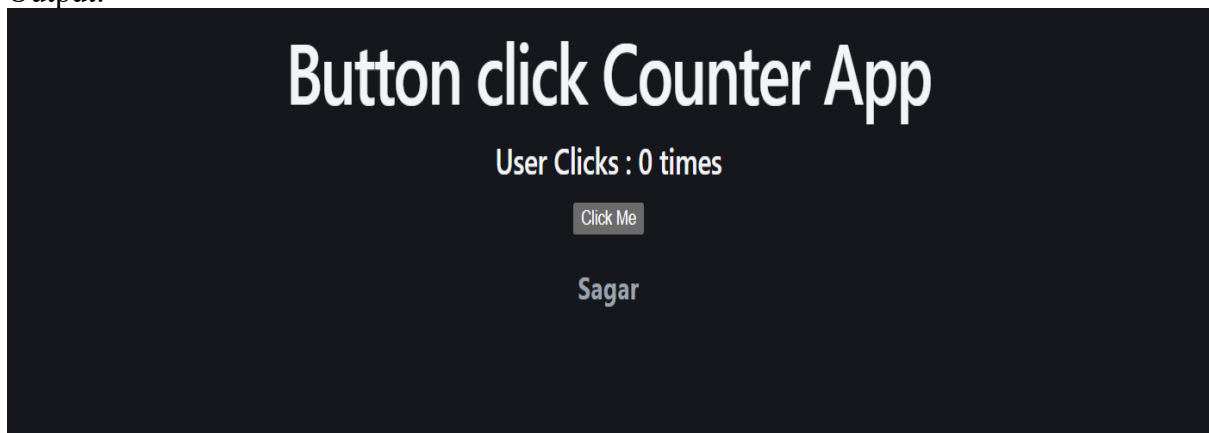
```
import {useState} from 'react';
function App() {
  const [count, setCount] = useState(0);
  return (

    <div className="App">

      <h1>Button click Counter App</h1>
      <h2>User Clicks : {count} times</h2>
      <button onClick={() => setCount(count + 1)}>Click Me</button>
      <h3>Sagar</h3>
    </div>
  );
}

export default App;
```

Output:



d. Create a form component to accept user input and display it dynamically.

Code:

StudentForm.jsx

```
import { useState } from "react";
```

```
function StudentRegisterForm() {  
  const [name, setName] = useState("");  
  const [email, setEmail] = useState("");  
  const [phone, setPhone] = useState("");  
  const [CGPA, setCgpa] = useState("");  
  
  return (  
    <div>  
      <h1>Student Registration Form</h1>  
      <hr/>  
      <label>Name:</label>  
      <input type="text" placeholder="Enter Name" value={name}  
onChange={(e) => setName(e.target.value)} />  
      <br/>  
      <label>Email:</label>  
      <input type="email" placeholder="Enter Email" value={email}  
onChange={(e) => setEmail(e.target.value)} />  
      <br/>  
      <label>Phone Number:</label>  
      <input type="text" placeholder="Enter Phone Number" value={phone}  
onChange={(e) => setPhone(e.target.value)} />  
      <br/>  
      <label>CGPA:</label>  
      <input type="text" placeholder="Enter CGPA" value={CGPA}  
onChange={(e) => setCgpa(e.target.value)} />  
      <br/>  
      <button type="submit" onClick={() => {}}>Submit</button>  
      <hr/>  
      <h2>Student details</h2>  
      <p>Name: {name}</p>  
      <p>Email: {email}</p>  
      <p>Phone Number: {phone}</p>  
      <p>CGPA: {CGPA}</p>  
    </div>  
  )  
}
```

Sagar Kandari
T010

```
);  
}
```

```
export default StudentRegisterForm;
```

```
App.jsx  
import StudentForm from "../components/StudentForm";
```

```
function App() {
```

```
  return (  
    <div>  
      <StudentForm ></StudentForm>  
  
    </div>  
  );  
}
```

```
export default App;
```

Output:

Student Registration Form

Name: Sagar

Email: sagarkandari8@gmail.com

Phone Number: 09987955278

CGPA: 9.0

Submit

Student details

Name: Sagar

Email: sagarkandari8@gmail.com

Phone Number: 09987955278

CGPA: 9.0

e. Display a list of items stored in the state using map().

Code:

App.jsx

```
import CompanyList from './components/CompanyList';  
import './App.css';
```

```
function App() {  
  return (  
    <div className="App">  
      <h1>Company List</h1>  
      <CompanyList />  
    </div>  
  );  
}
```

```
export default App;
```

Sagar Kandari
T010

CompanyList.jsx

```
import {useState} from 'react';

function CompanyList() {
  const [companies] = useState([
    "TCS",
    "Infosys",
    "Wipro",
    "HCL",
    "Tech Mahindra"
  ]);

  return (
    <div>
      <h2>Placement Companies</h2>
      <h3>Sagar</h3>
      <hr/>
      <ol>
        {companies.map((company, index) => (
          <li key={index}>{company}</li>
        ))}
      </ol>
    </div>
  );
}

export default CompanyList;
```

Output:

Company List	
Placement Companies	
Sagar	
1.	TCS
2.	Infosys
3.	Wipro
4.	HCL
5.	Tech Mahindra

f. Build a small interactive component (e.g., counter or to-do list).

Code:

App.jsx

```
import {useState} from 'react';
function App() {
  const [count, setCount] = useState(0);
  return (

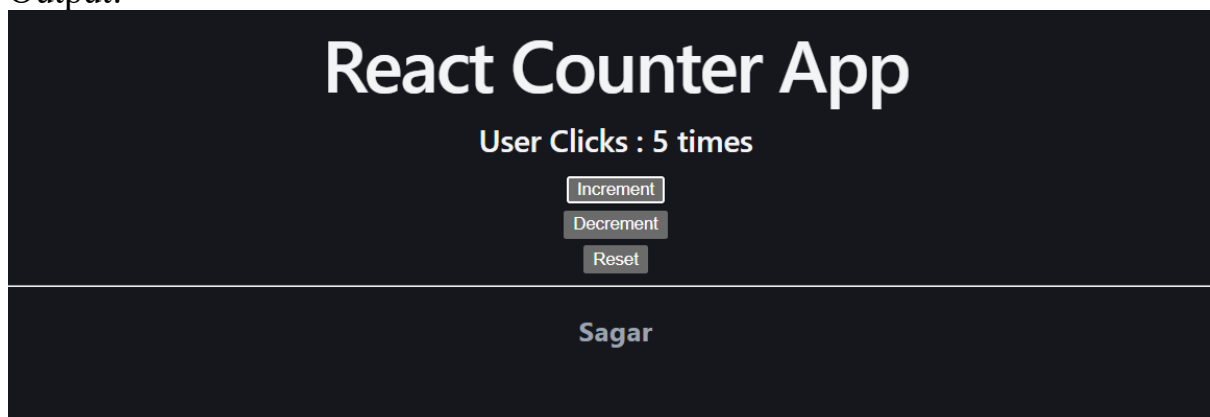
    <div className="App">

      <h1>React Counter App</h1>
      <h2>User Clicks : {count} times</h2>
      <button onClick={() => setCount(count + 1)}>Increment</button>
      <br/>
      <button onClick={() => setCount(count - 1)}>Decrement</button>
      <br/>
      <button onClick={() => setCount(0)}>Reset</button>
      <hr/>

      <h3>Sagar</h3>
    </div>
  );
}

export default App;
```

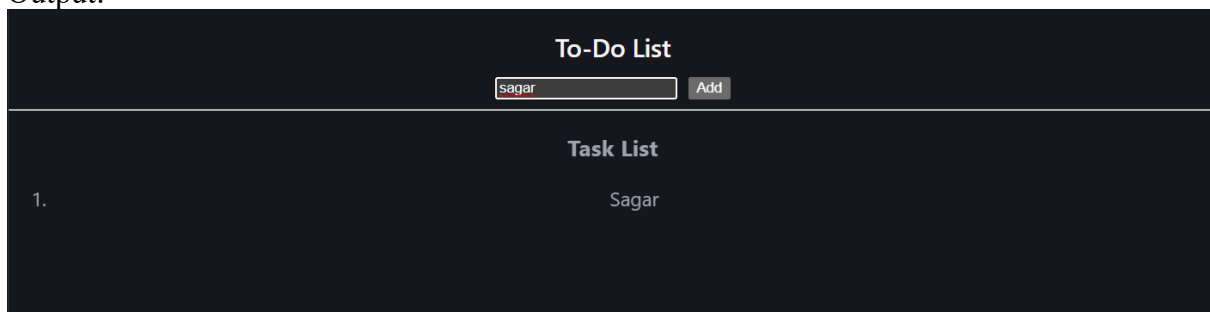
Output:



To do list:

```
import { useState } from "react";
function App() {
  const [task, setTask] = useState("");
  const [tasks, setTasks] = useState([]);
  // Add a new task
  const addTask = () => {
    if (task.trim() === "") {
      alert("Please enter a task!");
      return;
    }
    setTasks([...tasks, task]);
    setTask("");
  };
  return (
    <div style={{ textAlign: "center", marginTop: "30px" }}>
      <h2>To-Do List</h2>
      <input
        type="text"
        placeholder="Enter a task"
        value={task}
        onChange={(e) => setTask(e.target.value)}
      />
      <button onClick={addTask} style={{ marginLeft: "10px" }}>
        Add
      </button>
      <hr />
      <h3>Task List</h3>
      <ol>
        {tasks.map((item, index) => (
          <li key={index}>{item}</li>
        ))}
      </ol>
    </div>
  );
}
export default App;
```

Output:



The screenshot shows a web application titled "To-Do List". It features a text input field containing the word "sagar" and an "Add" button. Below the input is a horizontal line. Underneath the line is a section titled "Task List". This section contains a single list item: "1. Sagar".